

LOW TEMPERATURE MAGNETISM IN RARE-EARTH GARNETS: FROM MAGNETOCALORIC COOLING TO EFFECTIVE SPIN- $\frac{1}{2}$ PHYSICS

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MASTER OF SCIENCE

in **PHYSICS**

by

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to

SCHOOL OF PHYSICS
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April 2026

DECLARATION

I, Govind Unnikrishnan (Roll No: IMS21074), hereby declare that, this report entitled “**Low Temperature Magnetism in Rare-Earth Garnets: From Magnetocaloric Cooling to Effective Spin- $\frac{1}{2}$ Physics**” submitted to Indian Institute of Science Education and Research Thiruvananthapuram towards the partial requirement of **Master of Science in School of physics**, is an original work carried out by me under the supervision of **Dr. Deepshikha Jaiswal-Nagar** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold academic ethics and honesty. Whenever a piece of external information or statement or result is used then, that has been duly acknowledged and cited.

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14 April 2026

CERTIFICATE

This is to certify that the work contained in this project report entitled “Low Temperature Magnetism in Rare-Earth Garnets: From Magnetocaloric Cooling to Effective Spin- $\frac{1}{2}$ Physics” submitted by Govind Unnikrishnan (Roll No: IMS21074) to Indian Institute of Science Education and Research, Thiruvananthapuram towards the partial requirement of Master of Science in Physics has been carried out by him under my supervision and that it has not been submitted elsewhere for the award of any degree.

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ABSTRACT

This thesis presents a comprehensive investigation of rare earth garnet systems, focusing on both applied and fundamental aspects of low temperature magnetism. The first part explores gadolinium lithium tellurate garnet ($\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$, LGTO) as a candidate for adiabatic demagnetization refrigeration (ADR). Owing to its large magnetic entropy, weak magnetic anisotropy, and geometric frustration, LGTO exhibits excellent cooling performance, achieving temperatures as low as 165 mK with prolonged holding times, surpassing conventional benchmark materials. The second part investigates the sibling material lithium neodymium tellurate garnet ($\text{Li}_3\text{Nd}_3\text{Te}_2\text{O}_{12}$, LNTO) as a platform to realize an effective spin-1/2 ground state. Magnetic susceptibility and heat capacity measurements reveal a well isolated crystal electric field (CEF) ground state Kramers doublet, with a large energy gap to excited states. The reduced low temperature magnetic moment, field dependent Schottky anomaly, and absence of long range magnetic ordering indicate the emergence of $J_{\text{eff}} = 1/2$ behavior driven by strong spin-orbit coupling and geometric frustration. ADR measurements further demonstrate cooling down to 282 mK, reflecting persistent magnetic entropy at low temperatures.

Together, these results highlight the dual potential of rare earth garnets as efficient magnetic refrigerants and as model systems for exploring quantum magnetism in frustrated lattices with effective spin-1/2 ground state.

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Chapter 1

Introduction

Rare earth based magnetic systems are of considerable interest due to the unique properties arising from the localized nature of the $4f$ electrons. These electrons are well shielded by outer orbitals, leading to weak hybridization with the lattice and the retention of strong atomic character. As a result, rare earth ions exhibit well defined magnetic moments and a rich variety of ground states governed by the interplay of spin–orbit coupling, crystal electric field effects, and exchange interactions [2]. Depending on the electronic configuration, these systems can display large magnetic moments and high spin ground states, or strong spin-orbit coupling that lifts degeneracies and gives rise to low energy excitations and enhanced quantum fluctuations [2]. Such characteristics make rare earth compounds ideal platforms for both fundamental and applied research. On one hand, they host a variety of unconventional magnetic behaviors, while on the other, their large magnetic entropy and tunable interactions make them promising candidates for applications such as magnetic refrigeration, quantum materials, and advanced functional devices.

As mentioned above, rare earth magnetic systems exhibit a wide variety of ground states arising from the interplay of spin, orbital, and lattice degrees of freedom. The localized $4f$ electrons can display markedly different behavior depending on the relative contributions of spin and orbital angular momentum [2]. In particular, ions such as Gd^{3+} , with a half filled $4f$ shell ($S = 7/2$, $L = 0$), possess negligible orbital contribution and hence minimal spin-orbit coupling effects [3]. As a result, these systems exhibit nearly isotropic Heisenberg type exchange interactions with weak magnetic anisotropy [3]. The large spin value further leads to a high magnetic entropy ($R \ln(2S + 1)$), making such materials especially suitable for applications such as adiabatic demagnetization refrigeration where large and reversible entropy changes are essential for achieving ultra low temperatures.

Further, for some rare earth ions, strong spin–orbit coupling plays a dominant role. The orbital and spin angular momenta first couple to form well defined total angular momentum states denoted by J . The crystal electric field (CEF) subsequently lifts the $(2J + 1)$ fold degeneracy of this manifold, resulting in a series of discrete energy levels. For ions with an odd number of electrons, Kramers’ theorem ensures that each level remains at least doubly degenerate, forming Kramers doublets [4]. At sufficiently low temperatures, only the lowest lying doublet is popu-

lated, allowing the system to be effectively described as a spin-1/2 degree of freedom. Depending on the balance between interactions and lattice geometry, such systems can host a variety of ground states ranging from conventional magnetic order to highly correlated and quantum disordered phases. A schematic showing the evolution of effective spin - 1/2 state in Yb ions is shown in Fig. 1.1.

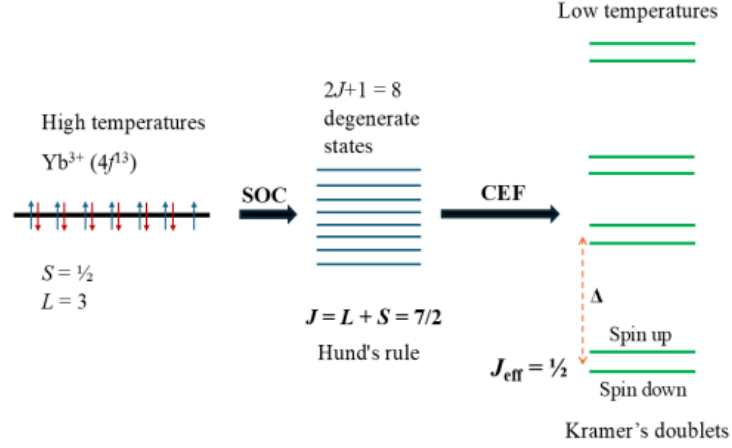


Figure 1.1: An example of crystal electric field splitting in a Yb based system [1]

Among the various crystal structures hosting rare earth ions, garnets are of particular interest due to their geometrically frustrated magnetic networks. In these systems, rare earth ions form a three dimensional arrangement of corner sharing triangles, commonly described as a hyperkagome lattice. This geometry leads to frustration, where competing interactions cannot be simultaneously satisfied, thereby suppressing long range magnetic ordering and preserving magnetic entropy to low temperatures [5]. A benchmark frustrated magnet for magnetocaloric studies in the sub Kelvin range is the most famous garnet $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ (GGG), belonging to the $\text{Gd}_3\text{T}_2\text{T}'_3\text{O}_{12}$ family [6, 7]. T and T' ions occupy the octahedral and tetrahedral sites, respectively and gives rise to three dimensional (3D) frustrated geometry through corner sharing network of polyhedrons. Thus, a strategic selection of T and T' ions provide the possibility of tuning magnetic interactions and consequently magnetocaloric performances. At the same time, in systems with strong spin-orbit coupling and crystal field effects, the same lattice can host effective spin-1/2 degrees of freedom, enabling the study of quantum magnetic behavior.

In this context, the present work investigates two garnet systems with complementary objectives. Gadolinium lithium tellurate garnet ($\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$, LGTO) is explored as a candidate for magnetic refrigeration, where the isotropic Gd^{3+} moments and frustrated lattice suppress magnetic ordering and retain entropy at low temperatures. In contrast, lithium neodymium tellurate garnet ($\text{Li}_3\text{Nd}_3\text{Te}_2\text{O}_{12}$, LNTTO) is studied to understand the role of crystal field effects and spin-orbit coupling in generating an effective $J_{\text{eff}} = 1/2$ ground state. For Nd^{3+} ions, the $J = 9/2$ multiplet is split into Kramers doublets, with the lowest doublet governing the low temperature behavior and giving rise to effective spin-1/2 physics in a frustrated lattice. Thus, this study connects both applied and fundamental aspects of rare earth magnetism.

Chapter 2

Experimental methodology and techniques used

Polycrystalline samples of $\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$ (LGTO) and $\text{Li}_3\text{Nd}_3\text{Te}_2\text{O}_{12}$ (LNTO) were synthesized using the conventional solid state reaction method. High purity starting materials, lithium carbonate (Li_2CO_3), tellurium dioxide (TeO_2), and the respective rare earth oxides (Gd_2O_3 / Nd_2O_3), were used in stoichiometric proportions. The powders were thoroughly mixed and homogenized using an agate mortar and pestle in an acetone medium for several hours to ensure uniformity. The mixture was then pelletized under uniaxial pressure and subjected to an initial heat treatment at 773 K for 5 hours. Subsequently, the pellets were reground, repelletized, and sintered at 1173 K for 9 hours. This process was repeated for 2-3 cycles to ensure phase homogeneity. Phase identification was carried out using powder X-ray diffraction (XRD) with a PANalytical diffractometer employing $\text{Cu-K}\alpha$ radiation. Data were collected in the 2θ range of 10° - 90° at room temperature. Rietveld refinement was performed using the FULLPROF software package to confirm the formation of the cubic garnet structure and to extract lattice parameters.

Magnetic measurements were performed using a Quantum Design Physical Property Measurement System (PPMS) equipped with a Vibrating Sample Magnetometer (VSM). Temperature dependent magnetization $M(T)$ and isothermal magnetization $M(H)$ measurements were carried out in the temperature range of 2-300 K under various applied magnetic fields for both compounds. In addition, heat capacity measurements were performed for LNTO to probe its low temperature magnetic behavior and crystal field effects. ADR measurements were carried out using a custom built setup to investigate the magnetocaloric properties of both compounds. The time evolution of temperature and magnetic field during the demagnetization process was recorded and analyzed. For LGTO, these measurements were primarily used to evaluate magnetocaloric performance, including estimation of the magnetic Grüneisen parameter and identification of field induced features. In the case of LNTO, ADR measurements were employed to explore the low temperature limit and entropy evolution associated with the effective spin-1/2 ground state.

Chapter 3

Frustrated Gd^{3+} garnets as candidates for ADR: The case of $\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$

Ultra low temperatures are essential for advancements in fields such as quantum computing, condensed matter physics, astrophysical detection, and precision metrology . Traditionally, such temperatures have been achieved using liquid helium based refrigeration [8]; however, the global helium shortage and the rising cost of isotopically pure ^3He have created a need for alternative cryogenic technologies. Among the most promising approaches are helium free, solid state magnetic refrigeration systems based on the magnetocaloric effect (MCE) or adiabatic demagnetization refrigeration (ADR), where cooling arises from the entropy change associated with spin alignment in an applied magnetic field [3]. Efficient materials for sub Kelvin refrigeration require a large magnetic entropy change ($-\Delta S_M$), low ordering temperatures, and weak spin correlations. In this context, Gd^{3+} -based compounds are particularly attractive due to their half-filled $4f$ shell ($S = 7/2$, $L = 0$), which leads to isotropic interactions, weak magnetic anisotropy, and large magnetic entropy, favoring strong magnetocaloric performance [5]. This first phase of work explores $\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$ (LGTO) as a possible ADR material.

3.1 Structural details

The room temperature powder XRD pattern of LGTO and the final refinement parameters obtained using the FULLProf software are shown in Fig. 3.1. The diffraction profile exhibits no impurity peaks, and all reflections match well with the standard pattern, confirming that LGTO is formed in a single phase with a cubic $Ia\bar{3}d$ space group. The CIF file obtained after phase identification was visualized using VESTA to examine the crystal structure, as shown in Fig. 3.2. The structure reveals a high density of magnetic sublattices, with Gd^{3+} ions occupying nearly all cationic sites. These ions form a three dimensional hyperkagome lattice of corner sharing triangles (Fig. 3.1(a)), leading to geometric frustration, where antiferromagnetic interactions cannot be simultaneously satisfied.

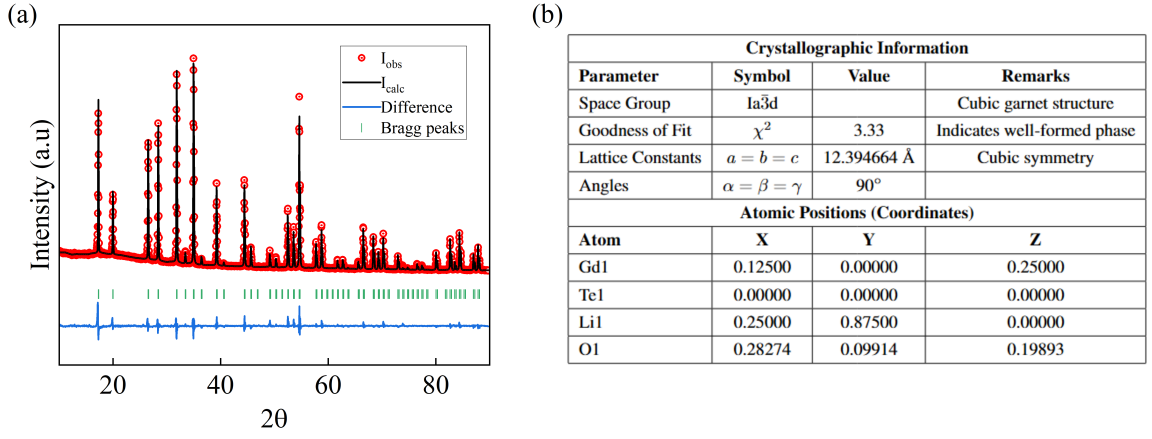


Figure 3.1: (a) Room temperature XRD pattern of LGTO along with the Rietveld refinement obtained using the FULLProf package. (b) Final structural refinement parameters corresponding to the best fit.

As a result, long range magnetic order is suppressed, giving rise to weak spin correlations over a wide temperature range. Furthermore, the large Gd-Gd separations and the presence of nonmagnetic Te^{6+} and Li^+ ions lead to weak superexchange interactions. Consequently, the exchange coupling (J) is expected to be small, and the magnetic energy scale is easily overcome by thermal fluctuations ($k_B T$), further suppressing magnetic ordering to very low temperatures.

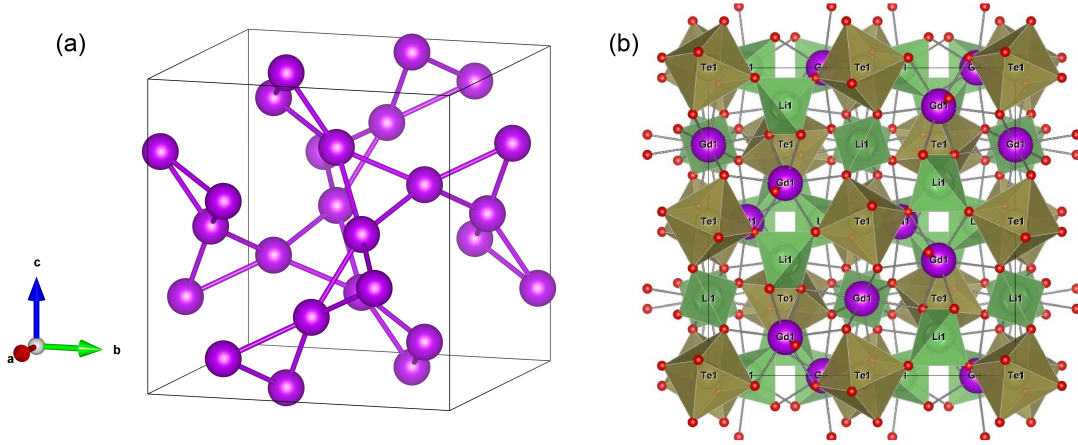


Figure 3.2: a) Magnetic Gd^{3+} ions located at the vertices of corner sharing triangles. b) Crystal structure of LGTO.

3.2 Magnetic properties

To characterize the magnetic ground state and verify the implications of the frustrated exchange network described above, temperature dependent magnetization measurements were performed under both zero-field-cooled (ZFC) and field-cooled (FC) protocols down to 1.8 K. The left panel of Fig.3.3 (a) shows the ZFC and FC susceptibility curves, which clearly indicate the absence of peak corresponding to magnetic ordering or spin glass like behaviour down to 1.8 K as there is no bifurcation between ZFC and FC curves.

To quantify the magnetic interactions, the inverse susceptibility was analyzed using the Curie-Weiss law. The fit (Fig. 3.3(a)) yields $\theta_{CW} = -2.50$ K and $C = 24.3$ emu K mol⁻¹. The corresponding effective moment, $\mu_{\text{eff}} = 2.828\sqrt{C}$, is $8.03 \mu_B$, in good agreement with the theoretical value of $7.94 \mu_B$. The negative θ_{CW} indicates dominant antiferromagnetic interactions. The absence of magnetic ordering near θ_{CW} suggests the presence of frustration arising from the hyper kagome lattice. Further, the exchange interaction parameter calculated using the following equation:

$$J_{ex} = \frac{3k_B\theta_{CW}}{zS(S+1)} = -0.059K,$$

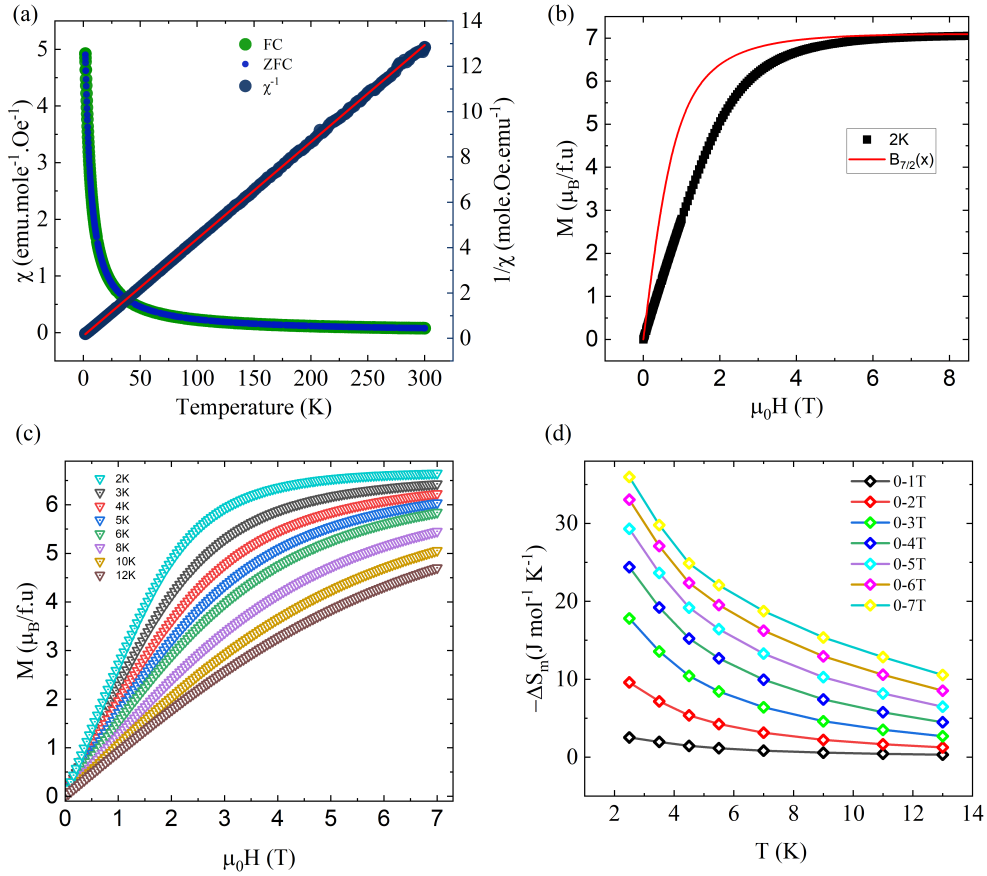


Figure 3.3: (a) Temperature variation of susceptibility (χ) and inverse susceptibility (χ^{-1}) with Curie-Weiss fitting (red line). (b) Isothermal magnetization versus applied field $M(H)$ at 2 K with Brillouin fitting, shown by solid red curve. (c) Isothermal magnetization curves $M(H)$ obtained at different temperatures in the range 2-20 K. (d) Magnetic entropy change ($-\Delta S_M$) for LGTO as a function of temperature in the range 2-12 K

is negative, suggesting weak antiferromagnetic interactions among the magnetic ions. Afterwards, magnetic field dependence of the magnetization was measured in order to probe the possibility of any kind of field induced behavior. The M - H isotherms reveal how the spins respond to an applied magnetic field when thermal fluctuations are minimal. Therefore, M - H was measured at 2 K, a temperature (less than $\theta_{CW} = -2.5K$) low enough to suppress thermal disorder, as shown in Fig. 3.3 (b). The 2 K isotherm was fitted using the Brillouin function,

which considers non interacting spins corresponding to a paramagnetic system. The fit matches well in the high field region but shows a noticeable deviation at low fields. Such deviations are typically associated with the development of antiferromagnetic correlations. The magnetization starts saturating at a field around 6 T. To further investigate the field and temperature dependence of the possible magnetic correlations, M – H isotherms at different temperatures are shown in Fig. 3.3 (c). It was found that as the temperature increases, the curves gradually approach Brillouin-like behaviour, consistent with dominant paramagnetic contributions.

For a quantitative understanding of the magnetocaloric behaviour, the magnetic entropy change ΔS_m was obtained using the relation:

$$\Delta S_m(T, H) = \int_0^H \left(\frac{\partial M}{\partial T} \right)_H dH.$$

The derivative $(\partial M/\partial T)_H$ was evaluated from the M – H isotherms measured in the first quadrant, and the magnetic entropy change $-\Delta S_m$ was calculated by integrating for different field intervals. The magnitude and temperature dependence of $-\Delta S_m$ provide insight into how efficiently the spins respond to an applied magnetic field. For low and intermediate fields, $-\Delta S_m$ decreases smoothly with temperature, consistent with predominantly paramagnetic behaviour governed by thermal disorder and weak antiferromagnetic correlations. However, at higher fields the $-\Delta S_m(T)$ curves begin to develop a more pronounced peak like behaviour, with the magnitude increasing rapidly. Such an evolution of the entropy profile indicate the field induced polarization of spins, where the applied magnetic field drives the system closer to a crossover or instability in the magnetic energy landscape. The non-saturating large value of $-\Delta S_m$ at the lowest temperature for 7 T field signifies large entropy accumulation at temperatures lower than 1.8 K, suggesting the possibility of achieving very low temperatures in the ADR test.

3.3 Adiabatic Demagnetization Refrigeration (ADR)

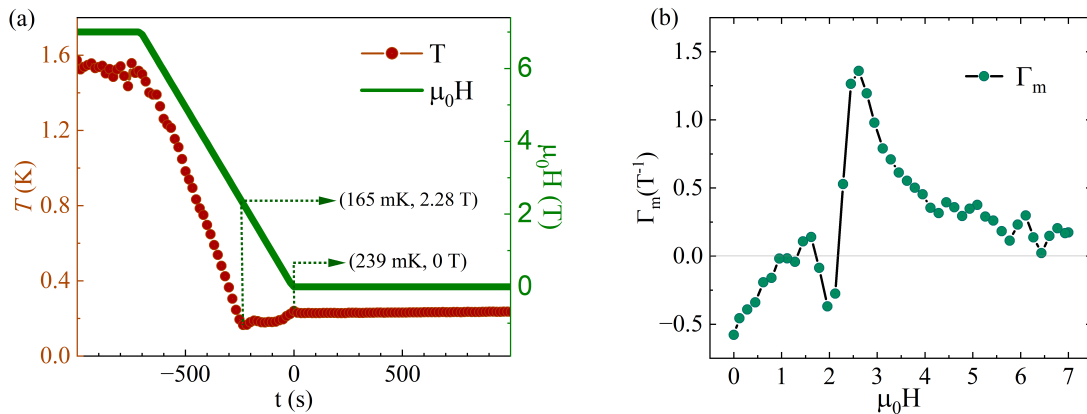


Figure 3.4: a) Left panel: sample temperature as a function of time; right panel: magnetic field variation with time b) Grüneisen parameter as a function of magnetic field

To directly measure the temperature change and holding time of LGTO, ADR measurements were performed by monitoring the variation of temperature and magnetic field as a function of time during a demagnetization cycle. The ADR data is shown in Fig. 3.4 which shows the time evolution of sample temperature as a function of time (in the left panel) when the magnetic field is adiabatically reduced (right panel). It reveals a minimum attainable temperature of approximately 165 mK at an applied field of 2.28 T and a holding time exceeding 12 hours. As the field reaches zero, the sample temperature slightly increases to 239 mK. This behaviour is consistent with a highly frustrated magnetic system, where the suppression of long range magnetic order to very low temperatures enables efficient adiabatic demagnetization. Notably, the fact that the lowest temperature is achieved at a finite field, rather than strictly at zero field, suggests that the system undergoes a field induced changes in the magnetic state. This further suggests the enhancement of entropy at the finite field of 2.28 T. In order to understand such behavior, magnetic Grüneisen parameter has proven to be a vital quantity. It measures the sensitivity of entropy to an external control parameter; in this case, the applied magnetic field. For a field-driven process, it is defined as:

$$\Gamma_{\text{mag}} = \frac{1}{T} \left(\frac{\partial T}{\partial H} \right)_S$$

which relates the change in temperature with field under adiabatic conditions. Divergence like behaviour or a sign change in Γ_{mag} is an indicator of field-induced transitions or proximity to a critical point. Fig. 5 (b) depicts the field dependence of Γ_{mag} evaluated from the ADR data and the observed sign change confirms the presence of field induced transition.

3.4 Conclusion

In conclusion, ADR measurements demonstrate that LGTO exhibits excellent cooling performance, achieving a minimum temperature of 165 mK with a prolonged holding time exceeding 12 hours below 1 K. This performance significantly surpasses that of benchmark refrigerants such as GGG and EuB_4O_7 under comparable conditions. The observed behavior can be attributed to enhanced geometric frustration, as inferred from structural analysis and magnetization measurements. Additionally, the possible presence of field induced critical points may further contribute to achieving such low temperatures with extended holding times, which will be investigated in detail through heat capacity measurements. These results establish LGTO as a highly promising candidate for sub Kelvin magnetic refrigeration and underscore its potential for practical ADR applications.

Chapter 4

Investigation of effective spin - 1/2 ground state in antiferromagnetic $\text{Li}_3\text{Nd}_3\text{Te}_2\text{O}_{12}$

Effective spin-1/2 systems, commonly referred to as ($J_{\text{eff}} = 1/2$) states, have attracted significant attention due to their ability to host a variety of exotic quantum phenomena such as quantum spin liquids and other strongly correlated states [9]. In rare earth systems, the presence of localized 4*f* electrons, strong spin-orbit coupling, and crystal electric field (CEF) effects provides a natural platform for realizing such states [4]. The spin and orbital angular momenta first couple to form well defined total angular momentum states, which are subsequently split by the crystal field into discrete energy levels. For ions with an odd number of electrons, Kramers' theorem ensures that these levels remain doubly degenerate, forming Kramers doublets [10]. At sufficiently low temperatures, only the lowest lying doublet is populated, allowing the system to be effectively described as a spin-1/2 degree of freedom. Garnet systems further provide a favorable setting for such physics due to their geometrically frustrated hyperkagome lattice of corner sharing triangles, which suppresses long range magnetic ordering and enhances quantum fluctuations. Efficient realization of ($J_{\text{eff}} = 1/2$) behavior in such lattices offers an opportunity to explore unconventional magnetic ground states and low-energy excitations. In this context, the second phase of this work investigates $\text{Li}_3\text{Nd}_3\text{Te}_2\text{O}_{12}$ (LNTO) as a candidate system to realize and study an effective ($J_{\text{eff}} = 1/2$) ground state.

4.1 Structural details

The room temperature powder XRD pattern of LNTO and the final refinement parameters obtained using the FULLProf software are shown in Fig. 4.1. The diffraction profile exhibits no impurity peaks, and all reflections match well with the standard pattern, confirming that LNTO is formed in a single phase with a cubic $Ia\bar{3}d$ space group. The crystal structure of LNTO obtained from CIF analysis and visualized using VESTA, reveals that the Nd^{3+} ions form a three dimensional hyperkagome lattice similar to its Gd analogue.

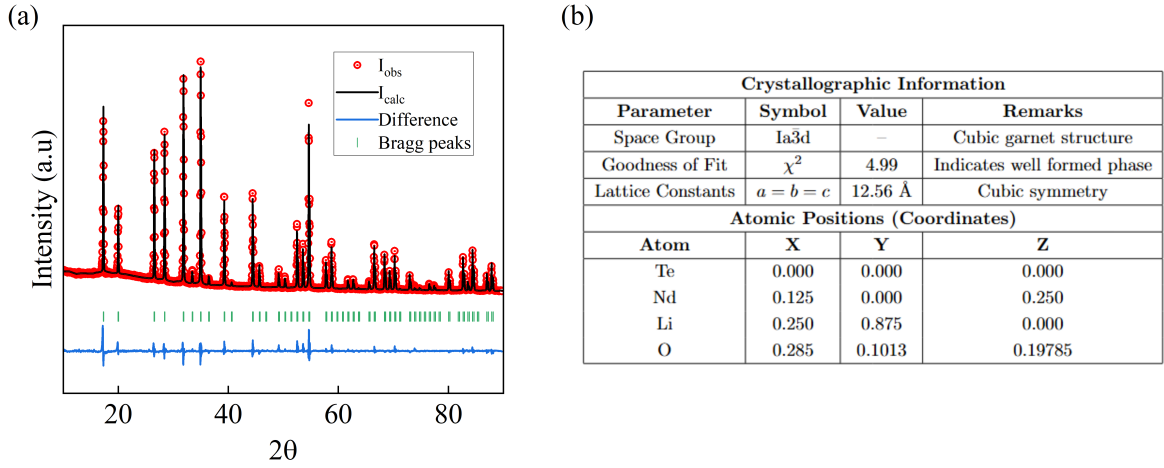


Figure 4.1: (a) Room temperature XRD pattern of LGTO along with the Rietveld refinement obtained using the FULLProf package. (b) Final structural refinement parameters corresponding to the best fit.

In this structure, the magnetic Nd ions occupy a network of corner sharing triangles, giving rise to strong geometric frustration. However, in contrast to LGTO, where the Gd^{3+} ions exhibit a large spin ($S = 7/2$) with nearly isotropic Heisenberg type interactions, the Nd^{3+} ions in LNT0 experience strong crystal field (CEF) effects and spin-orbit coupling (SOC), which together can generate an effective spin 1/2 ground state at low temperatures.

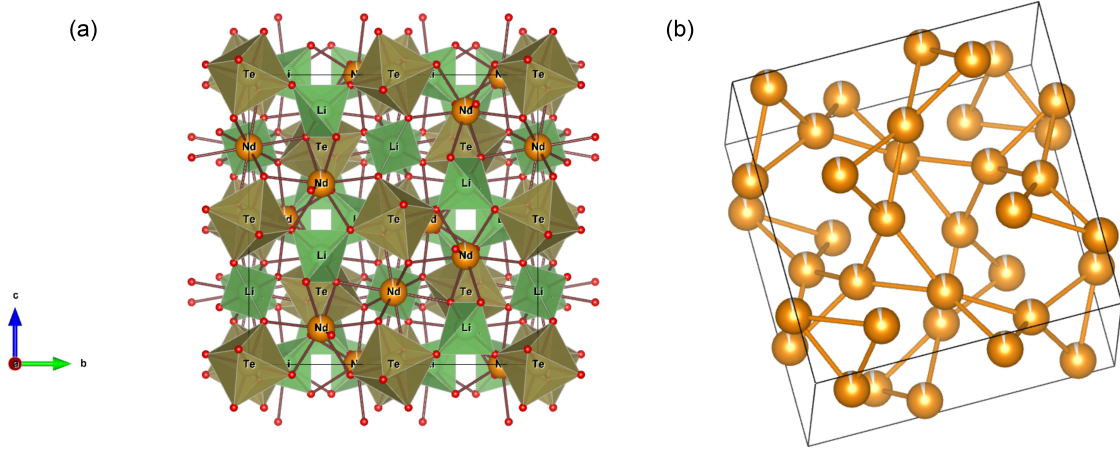


Figure 4.2: a) Crystal structure of LNT0 b) Magnetic Nd^{3+} ions located at the vertices of corner-sharing triangles.

4.2 Magnetic properties

Temperature dependence of the DC magnetic susceptibility (χ_{DC}) of LNT0 was measured under zero-field-cooled (ZFC) and field-cooled (FC) conditions at 100 Oe in the range 1.8–300 K. No anomaly is observed down to 1.8 K, indicating the absence of long range magnetic ordering. The ZFC and FC curves overlap over the entire temperature range (Fig. 4.3(a)), ruling out spin freezing or glassy behavior. The inverse susceptibility (χ_{DC}^{-1}) shows linear behavior in the high temperature (HT) regime, with a deviation at low temperatures due to crystal field (CF) effects. The HT data were fitted using the Curie-Weiss law, yielding $C_{\text{HT}} = 5.344 \pm 0.418 \text{ emu/mol-Oe-K}$

and $\theta_{\text{HT}} = -63.96 \pm 10.33$ K. The large negative θ_{HT} suggests antiferromagnetic correlations, although its magnitude is enhanced by contributions from thermally populated excited CF levels. Normalizing per Nd^{3+} ion, the effective moment is $\mu_{\text{eff}}^{\text{HT}} \approx 3.78 \mu_B$, close to the free-ion value ($\approx 3.62 \mu_B$), indicating that all CF levels contribute in this regime.

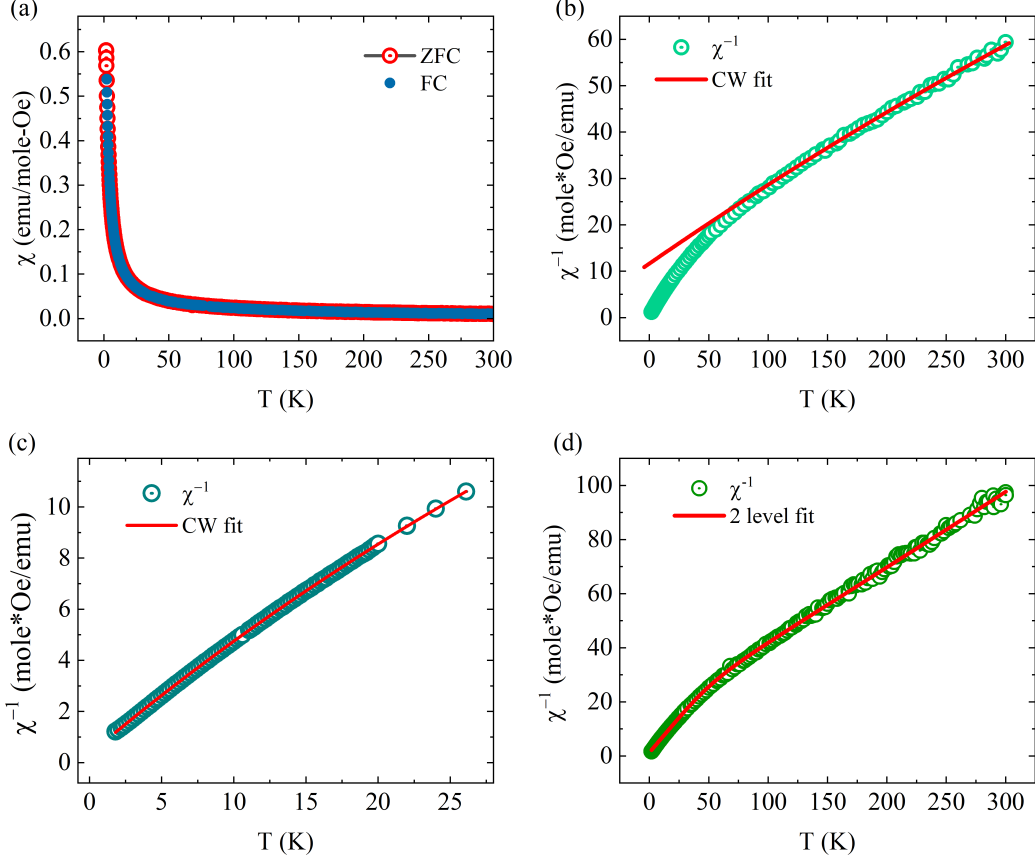


Figure 4.3: (a) Temperature dependence of magnetic susceptibility (χ) under ZFC and FC conditions. (b) Temperature dependence of inverse susceptibility (χ^{-1}) with Curie-Weiss fitting in the high temperature regime. (c) Temperature dependence of inverse susceptibility (χ^{-1}) with Curie-Weiss fitting in the low temperature regime. (d) Temperature dependence of inverse susceptibility (χ^{-1}) fitted using the two-level fitting.

To probe the intrinsic low temperature behavior, data below 10 K were fitted using a modified Curie-Weiss expression, yielding $C_{\text{LT}} = 2.072 \pm 0.004$ emu/mol-Oe-K and $\theta_{\text{LT}} = -0.690 \pm 0.011$ K. The corresponding effective moment per Nd^{3+} ion is $\mu_{\text{eff}}^{\text{LT}} \approx 2.35 \mu_B$, significantly reduced compared to the HT value. This reduction indicates that only the crystal field ground state Kramers doublet is populated, and the system can be effectively described by a $J_{\text{eff}} = 1/2$ model. For Nd^{3+} ($4f^3$), strong spin-orbit coupling leads to a total angular momentum $J = 9/2$, with the $2J + 1 = 10$ manifold split into five Kramers doublets by the crystal field. At low temperatures, only the ground state doublet is populated, resulting in the observed reduction in effective moment and the emergence of effective spin-1/2 behavior.

The small negative value of θ_{LT} indicates very weak antiferromagnetic interactions among the Nd^{3+} ions, consistent with the localized nature of the $4f$ orbitals and weak superexchange

pathways mediated through nonmagnetic ions. The suppression of long range magnetic ordering down to low temperatures can be attributed to the combined effects of weak exchange interactions, geometric frustration arising from the hyperkagome lattice, and enhanced quantum fluctuations associated with the effective spin-1/2 ground state.

To further understand the influence of excited CF states on the lowest kramers doublet, the inverse susceptibility was analysed using a two level model [9],

$$\chi_{\text{DC}}^{-1} = 8(T - \theta_{\text{CW}}) \left(\frac{\mu_{\text{eff},1}^2 + \mu_{\text{eff},2}^2 e^{-\Delta_{\text{CF}}/T}}{1 + e^{-\Delta_{\text{CF}}/T}} \right)^{-1}$$

where Δ represents the energy gap between the ground and first excited CF states. The fitting yields $\Delta = 164.71 \pm 3.44$ K and $\theta_{\text{CW}} = -2.43 \pm 0.20$ K. The large value of Δ indicates a substantial crystal field gap, ensuring that the ground state doublet is well isolated at low temperatures. This validates the description of LNTO as an effective spin-1/2 system in the low temperature regime.

Additionally, isothermal magnetization $M(H)$ is measured at 2 K ($\gg \theta_{\text{LT}}$) and 300 K, as shown in Fig. 2(c). The magnetization curve at 2 K exhibits a clear deviation from linearity at higher magnetic fields, indicating the onset of saturation. This behaviour is analyzed using the Brillouin function,

$$\frac{M}{M_s} = \tanh \left(\frac{g_J J \mu_B H}{k_B T} \right),$$

where M_s is the saturation magnetization. The fitting yields a Landé g -factor $g_J \approx 2.17$ for $J = 1/2$.

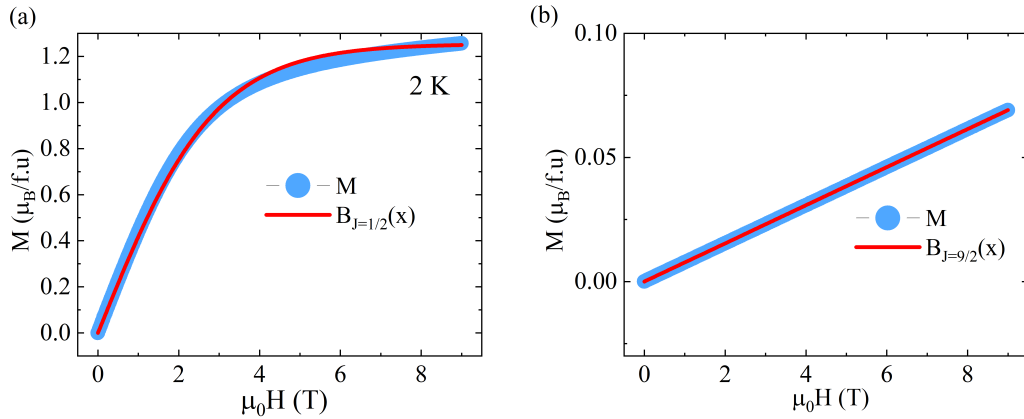


Figure 4.4: *Isothermal magnetization data at a) 2K and b) 300K*

In contrast, the magnetization measured at 300 K shows a linear dependence on the applied magnetic field, characteristic of paramagnetic behaviour. At such high temperatures, multiple crystal field levels are thermally populated, and the magnetization cannot be described by a single Brillouin function with a well defined J value. The apparent agreement with an effective J value in fitting arises from a phenomenological approximation rather than a true physical

state. The isothermal magnetization data at 300 K is well described by a Brillouin function with $J = 9/2$, consistent with the full multiplet of Nd^{3+} ions being thermally populated. In contrast, the 2 K data is well fitted with $J = 1/2$, indicating that only the ground state Kramers doublet is active at low temperatures. This reflects a crossover from a high temperature regime governed by the full $J = 9/2$ multiplet to a low temperature regime described by an effective $J_{\text{eff}} = 1/2$ ground state due to crystal field splitting.

4.3 Heat capacity

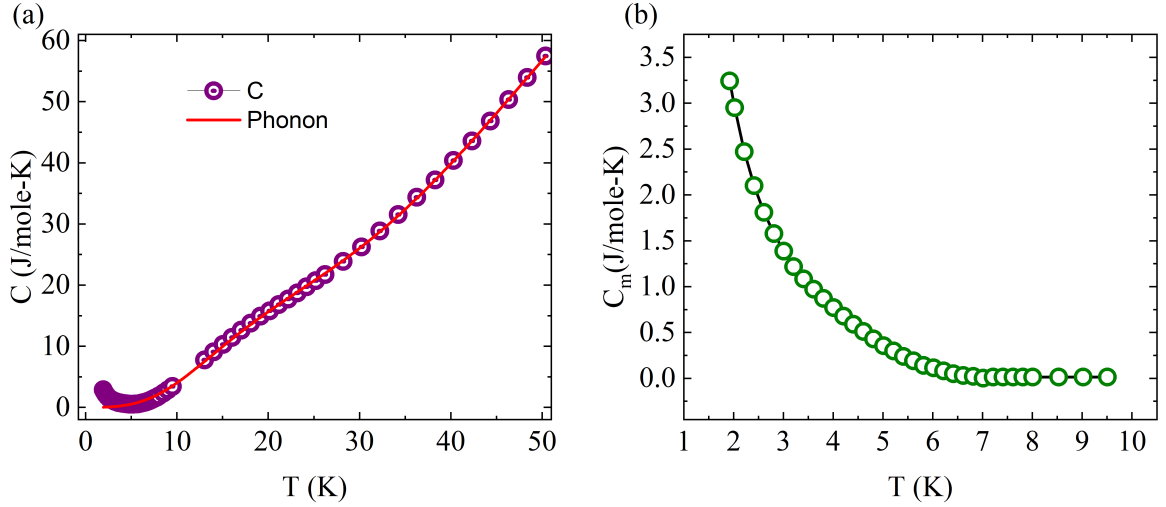


Figure 4.5: (a) Temperature dependence of the zero-field heat capacity fitted using the Debye–Einstein model to estimate the phonon contribution. (b) Magnetic heat capacity, $C_{\text{mag}}(T)$, obtained after subtracting the lattice (phonon) contribution from the total heat capacity.

To probe the low temperature thermodynamic properties of LNTO, temperature dependent heat capacity measurements, $C(T)$, are performed down to 1.8 K. The total heat capacity consists of contributions from lattice vibrations and possible magnetic degrees of freedom. In order to isolate the lattice contribution, the zero field heat capacity data is analyzed using a combined Debye–Einstein model, which effectively captures both acoustic and optical phonon contributions. The heat capacity is expressed as a linear combination of Debye and Einstein terms:

$$C(T) = m C_{\text{Debye}}(T) + (1 - m) C_{\text{Einstein}}(T),$$

where m represents the weight factor of the Debye contribution [10]. The Debye term accounts for low energy acoustic phonon modes, while the Einstein term represents optical phonon modes associated with localized vibrations. The Debye and Einstein contributions are given by:

$$C_{\text{Debye}}(T) = -9nR \left(\frac{T}{\theta_D} \right)^3 \int_{\theta_D/T}^0 \frac{x^4 e^x}{(e^x - 1)^2} dx$$

$$C_{\text{Einstein}}(T) = 3nR \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2},$$

where n is the number of atoms per formula unit, R is the universal gas constant, and θ_D and θ_E are the Debye and Einstein temperatures, respectively. This combined model provides a reliable description of the phononic background, enabling further analysis of low temperature magnetic contributions. The zero field heat capacity data in the high temperature region is fitted using the combined Debye-Einstein model, and the estimated phonon contribution is subtracted from the total heat capacity to extract the magnetic contribution, $C_{\text{mag}}(T)$. The obtained $C_{\text{mag}}(T)$ at zero field shows an upturn at low temperatures, suggesting a tendency towards a peak like feature. Such behaviour may arise either from a magnetic phase transition or from a Schottky type anomaly associated with low energy excitations.

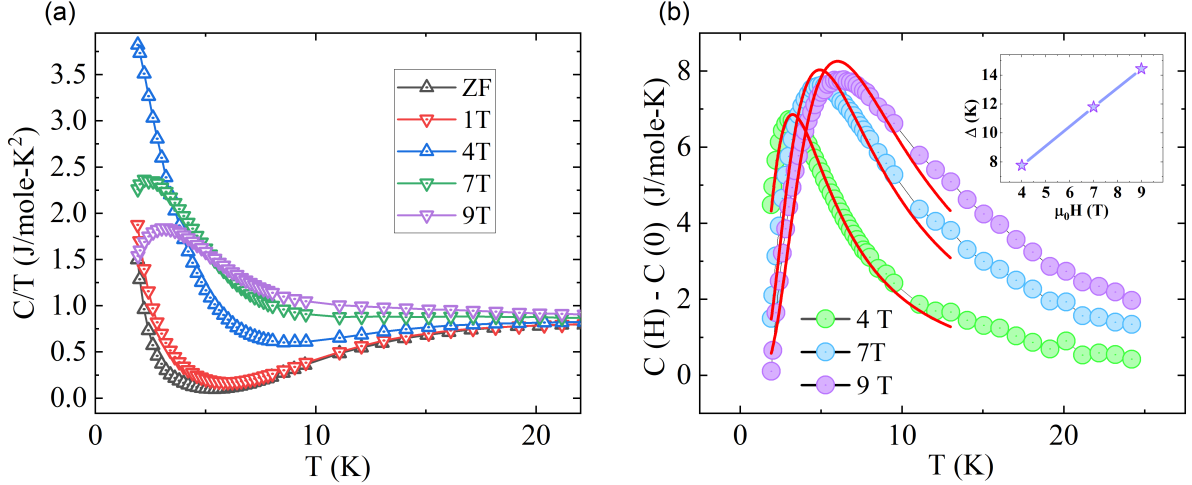


Figure 4.6: (a) Temperature dependence of C/T under various applied magnetic fields, showing the evolution and field induced shift of the low temperature Schottky anomaly. (b) Field induced heat capacity contribution, $C(H) - C(0)$, fitted using a two level Schottky model to extract the energy gap associated with Zeeman splitting.

To distinguish between these possibilities, the field dependence of the heat capacity is examined. In the case of a magnetic phase transition, the peak position typically exhibits weak dependence on the applied magnetic field, whereas a Schottky anomaly shows a pronounced field dependence. Therefore, the variation of C/T as a function of temperature under different applied magnetic fields is analyzed. Upon the application of external magnetic fields, a broad peak becomes clearly visible which systematically shifts towards higher temperatures with increasing field strength. This behavior is characteristic of a Schottky anomaly. With increasing magnetic field, both the ground and excited Kramers doublets undergo Zeeman splitting, leading to a modification of the energy level scheme. As a result, the effective energy gap between the thermally populated levels increases. Consequently, higher thermal energy is required to excite the system, which leads to a systematic shift of the Schottky peak towards higher temperatures with increasing magnetic field. To further understand this behavior, the magnetic contribution is analyzed using a two level Schottky model given by

$$C(H) - C(0) = N_A k_B \frac{(\Delta/T)^2 e^{-\Delta/T}}{(1 + e^{-\Delta/T})^2},$$

where Δ represents the Zeeman energy gap between the two levels of the split ground state doublet. The quantity $C(H) - C(0)$ isolates the field induced magnetic contribution. The experimental data for $C(H) - C(0)$ is fitted using the above expression, and the extracted values of Δ at different magnetic fields are found to follow a linear dependence on the applied field as shown in inset 4.6. This behavior confirms that the energy splitting arises from Zeeman interaction and can be described by $\Delta = g\mu_B H$. From the slope of the linear fit, the effective Landé g -factor is obtained as $g \approx 1.98$, which is in good agreement with the value obtained from low temperature magnetization measurements. Thus, the heat capacity results, in conjunction with magnetic studies, establish LNTO as a promising candidate for investigating the physics associated with effective spin-1/2 systems in three dimensional frustrated lattices.

4.4 Adiabatic Demagnetization Refrigeration (ADR)

To further probe the low temperature properties of LNTO, adiabatic demagnetization refrigeration (ADR) measurements were performed. In effective $J_{\text{eff}} = 1/2$ systems, the presence of a Kramers doublet provides a finite magnetic entropy of $R \ln 2$. In conventional magnetic systems, this entropy is typically released through long range magnetic ordering at low temperatures. However, in systems with strong quantum fluctuations and possible geometric frustration, the establishment of long range order is suppressed. As a result, the magnetic entropy is retained down to much lower temperatures, which enhances the efficiency of adiabatic demagnetization cooling.

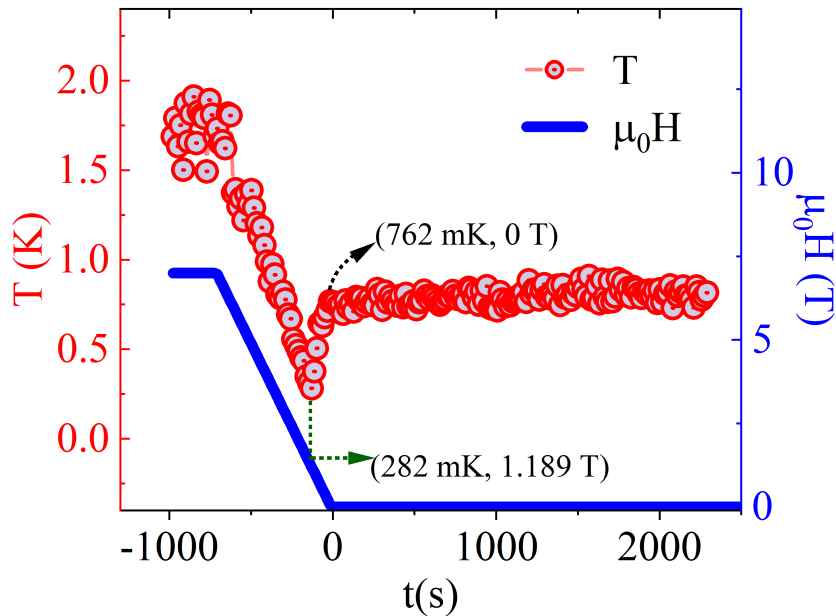


Figure 4.7: Left panel: Time dependence of the sample temperature; right panel: time dependence of the applied magnetic field

The ADR process begins by cooling the sample to a base temperature of approximately

1.8 K using a conventional ^4He cryostat. Under these conditions, thermal equilibrium between the sample and its surroundings is ensured by the presence of exchange gas, which provides efficient thermal contact and establishes isothermal conditions. A strong external magnetic field is then applied, causing the magnetic moments (spins) to align along the field direction, thereby reducing the magnetic entropy of the system. Subsequently, the exchange gas is evacuated, thermally isolating the sample and establishing adiabatic conditions. Once isolation is achieved, the magnetic field is gradually reduced. During this adiabatic demagnetization process, the total entropy remains constant. As the field decreases, the spins become increasingly disordered, leading to an increase in magnetic entropy. To compensate, the lattice (phonon) entropy decreases, resulting in a reduction of the sample temperature. The ultralow temperatures achieved in this process are measured using ruthenium oxide sensors integrated into a custom built ADR puck. Through this technique, temperatures well below 1 K can be reached, enabling the determination of the lowest temperatures accessible in LNTO.

The ADR data shown in Fig. 4.7 reveal that a minimum temperature of 282 mK is achieved at a finite magnetic field of 1.189 T, while at zero field the sample temperature reaches 762 mK. Notably, the experimentally reached temperatures are significantly lower than the characteristic temperature scale set by the Curie-Weiss temperature, θ_{CW} . In conventional magnetic systems, θ_{CW} provides an estimate of the interaction strength and the expected onset of magnetic correlations or ordering. However, in the present case, the system remains disordered down to temperatures well below this scale.

This suppression of long range magnetic ordering can be attributed to strong quantum fluctuations associated with the effective $J_{\text{eff}} = 1/2$ moments, as well as possible geometric frustration in the lattice. These effects prevent the complete release of magnetic entropy through ordering, allowing a substantial fraction of entropy to persist to very low temperatures. As a result, the system remains highly responsive to adiabatic demagnetization, enabling cooling to sub Kelvin temperatures which is an application of effective spin half systems.

4.5 Conclusion

In conclusion, the combined magnetic, thermodynamic, and ADR measurements establish LNTO as an effective spin-1/2 system with a well isolated Kramers doublet ground state. The absence of long range magnetic ordering, together with field dependent Schottky behavior and reduced low temperature moment, highlights the role of crystal field effects, geometric frustration, and quantum fluctuations. The ability to reach sub Kelvin temperatures through ADR further demonstrates the persistence of magnetic entropy, making LNTO a promising platform for exploring low temperature quantum magnetism in frustrated lattices.

Chapter 5

Future Direction

Future work will focus on a more detailed mapping of these field-induced transitions, locating the associated critical points, and probing the system at even lower temperatures to explore the possible existence of quantum critical end points in LGTO.

To further investigate the crystal field scheme of LNTO, the sample has been sent to the MARI time of flight inelastic neutron scattering (INS) spectrometer at ISIS Neutron and Muon Source for detailed measurements. INS provides a direct probe of the magnetic excitation spectrum, enabling the determination of crystal field excitations, spin dynamics, and possible signatures of fractionalized excitations associated with quantum magnetic states.

In particular, the INS measurements will help to establish the crystal electric field (CEF) level scheme of the Nd^{3+} ions and confirm the presence of an effective $J_{\text{eff}} = 1/2$ ground state. These measurements will complement the thermodynamic results, such as susceptibility, heat capacity, and ADR data, providing a comprehensive understanding of the low temperature magnetic behavior of LNTO.

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